

UNIT -3

1. Write a short note on statistics and methods used for data science.

Answer:

Statistics is the branch of mathematics concerned with the **collection, organization, presentation, analysis, and interpretation of data**. It plays a fundamental role in understanding patterns, making decisions, and drawing meaningful conclusions from data.

In the field of **data science**, statistics serves as the backbone for analyzing large datasets and ensuring that insights are accurate and reliable.

Methods Used in Data Science

1. **Data Collection**

Data is gathered from various sources such as surveys, experiments, databases, or online platforms. It can be primary or secondary data.

2. **Data Cleaning (Preprocessing)**

Raw data is processed to remove errors, missing values, and inconsistencies to ensure accuracy in analysis.

3. **Descriptive Statistics**

This involves summarizing data using measures such as **mean, median, mode, and standard deviation**, helping to understand the basic features of the dataset.

4. **Inferential Statistics**

It is used to make predictions or generalizations about a population based on a sample. It includes **hypothesis testing and confidence intervals**.

5. **Data Visualization**

Data is represented using charts and graphs such as **bar charts, pie charts, and histograms** to make interpretation easier.

6. **Probability Theory**

It deals with the study of uncertainty and likelihood of events, which is essential for making predictions.

7. **Machine Learning Techniques**

These involve algorithms that learn from data to make predictions or decisions, such as **regression and classification models**.

Conclusion

Statistics is an essential component of data science, providing the necessary tools and techniques to analyze data effectively and support informed decision-making.

2. What are the main methods of collecting primary data? State briefly the advantages and disadvantages of each.

Answer:

Primary data is data collected **first-hand by the investigator** for a specific purpose.

Methods of Collecting Primary Data (Easy Format)

1. Direct Personal Investigation

👉 Investigator collects data directly from people

Advantages:

- Accurate
- Reliable

Disadvantages:

- Time-consuming
 - Costly
-

2. Indirect Oral Investigation

👉 Data collected from third persons (witnesses/experts)

Advantages:

- Useful when direct data not possible
- Quick

Disadvantages:

- Less reliable
 - Chances of bias
-

3. Through Correspondents

👉 Local agents send data regularly

Advantages:

- Covers large area
- Cheap

Disadvantages:

- Less control
 - Errors possible
-

4. Mailed Questionnaire

👉 Questions sent (post/email), respondents fill

Advantages:

- Low cost
- Wide reach

Disadvantages:

- Low response
 - No clarification
-

5. Schedule Method (Enumerator)

👉 Investigator sends trained person to collect data

Advantages:

- More accurate
- Good response

Disadvantages:

- Expensive
 - Time-taking
-

Conclusion (1 line)

Choice depends on **cost, time, and accuracy required.**

✅ Memory Tip:

👉 **Direct → Indirect → Correspondent → Questionnaire → Schedule**
("D I C Q S")

Q3. Write a short note on primary and secondary data in statistics with examples.

Answer:

Primary Data

Primary data is the data that is **collected directly by the investigator** for a specific purpose. It is original and collected for the first time.

Examples:

- Conducting a survey to know students' study habits
- Collecting data through experiments or interviews

Advantages:

- More accurate and reliable
- Specific to the study

Disadvantages:

- Time-consuming
 - Expensive
-

Secondary Data

Secondary data is the data that has **already been collected by someone else** and is used by the investigator.

Examples:

- Government reports (Census data)
- Books, journals, websites

Advantages:

- Saves time and cost
- Easily available

Disadvantages:

- May not be accurate
 - May not suit the current purpose
-

Conclusion

Primary data is **original and reliable but costly**, whereas secondary data is **quick and economical but may lack accuracy**.

4. What do you understand by measure of dispersion? What purpose does a measure of dispersion serve?

Answer:

Measure of Dispersion – Definition

A **measure of dispersion** shows the **spread or variation of data values** around a central value (mean, median, or mode).

It tells us **how much the data is scattered or concentrated**.

Examples of Measures of Dispersion

- Range
 - Quartile Deviation
 - Mean Deviation
 - Standard Deviation
-

Purpose of Measure of Dispersion

1. **Shows Variability**
Helps to understand how much data values differ from each other.
 2. **Checks Reliability of Average**
If dispersion is low → average is reliable
If dispersion is high → average is less reliable
 3. **Comparison of Data Sets**
Helps compare consistency between two or more datasets.
 4. **Understanding Distribution**
Shows whether data is tightly grouped or widely spread.
 5. **Useful in Decision Making**
Helps in fields like business, economics, and research.
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Conclusion

Measures of dispersion are important because they **complement averages and give a complete understanding of data behavior**.

5. What are the measures of central tendency? Explain each.

Answer:

Measures of Central Tendency – Definition

Measures of central tendency are values that **represent the center or typical value of a dataset**. They help to summarize large data into a single value.

Main Measures of Central Tendency

1. Arithmetic Mean (Mean)

👉 The **average** of all observations

Formula:

Mean = (Sum of all values) / (Number of values)

Explanation:

It is the most commonly used measure and considers all data values.

Merits:

- Easy to calculate
- Uses all observations

Demerits:

- Affected by extreme values (outliers)
-

2. Median

👉 The **middle value** when data is arranged in order

Explanation:

- If odd number of values → middle value
- If even → average of two middle values

Merits:

- Not affected by extreme values
- Suitable for skewed data

Demerits:

- Does not use all observations

3. Mode

👉 The value that occurs **most frequently**

Explanation:

It shows the most common value in the dataset.

Merits:

- Easy to understand
- Useful for categorical data

Demerits:

- May not be unique
 - Not suitable for all datasets
-

Conclusion

Mean, median, and mode are the main measures of central tendency. Each has its own advantages and is used depending on the nature of the data.

Q6. List any applications of data science.

Answer:

Data science is widely used in various fields to analyze data and make better decisions. Some important applications are:

Applications of Data Science

1. Healthcare

- Disease prediction
- Patient data analysis
- Drug discovery

2. Business and Marketing

- Customer behavior analysis
- Sales forecasting
- Targeted advertising

3. Finance and Banking

- Fraud detection

- Risk analysis
- Credit scoring

4. **E-commerce**

- Product recommendations
- Personalized shopping experience
- Demand prediction

5. **Education**

- Student performance analysis
- Personalized learning

6. **Transportation**

- Traffic prediction
- Route optimization

7. **Social Media**

- Trend analysis
- Content recommendation

8. **Weather Forecasting**

- Climate prediction
- Disaster management

Conclusion

Data science is used in many real-life fields to **analyze data, improve efficiency, and support decision-making.**

14 Q. Define: (i) Population and Sample (ii) Sampling Distribution

(i) Population and Sample

Population

Population is the **entire collection of all observations or items** that we want to study.

👉 It includes all possible data related to a problem.

Example:

All students in a college

Sample

Sample is a **subset (small part) of the population** selected for study.

👉 It represents the population.

Example:

50 students selected from the college

(ii) Sampling Distribution

Sampling distribution is the **probability distribution of a statistic** (like mean) obtained from **all possible samples of a given size** from a population.

👉 It shows how sample values vary.

Example:

If we take many samples and calculate their means, the distribution of those means is called sampling distribution.

Conclusion

- Population → whole data
- Sample → part of population
- Sampling distribution → distribution of sample statistics

15 Q. Explain Point Estimation and Interval Estimation

Estimation – Meaning

Estimation is the process of using **sample data to estimate unknown population parameters** (like mean, proportion).

(i) Point Estimation

Definition

Point estimation is a method of estimating a population parameter by using a **single value**.

👉 It gives only **one exact number**.

Example

If sample mean = 50

👉 Then population mean ≈ 50

(This single value is the estimate)

Features

- Simple and easy
 - Quick to calculate
 - Less information (no accuracy range)
-

(ii) Interval Estimation

Definition

Interval estimation gives a **range of values** within which the population parameter is expected to lie.

👉 It includes **confidence level** (like 95%)

Example

Population mean lies between **48 and 52** with 95% confidence

Features

- More reliable
- Gives range instead of single value
- Includes level of confidence

Difference

Point Estimation	Interval Estimation
Single value	Range of values
Less reliable	More reliable
No confidence level	Includes confidence level

Conclusion

Point estimation gives a **single estimate**, while interval estimation provides a **range with confidence**, making it more useful in practical situations.

16 Q. Define Population and Sample with examples

Population

Definition:

Population is the **entire set of all observations or items** that are of interest in a statistical study.

👉 It includes all possible data.

Examples:

- All students in a university
 - All employees in a company
 - All households in a city
-

Sample

Definition:

Sample is a **small subset of the population** selected for analysis.

👉 It represents the population.

Examples:

- 100 students selected from a university
 - 50 employees chosen from a company
 - 200 households surveyed from a city
-

Conclusion

Population is the **whole group**, while a sample is a **part of that group** used for study.

Q24. Define population and sample. Explain various sampling methods.

1. Population

Population is the **complete set of all observations or items** under study.

👉 Example:
All students in a college

2. Sample

Sample is a **subset of the population** selected for analysis.

👉 Example:
100 students selected from a college

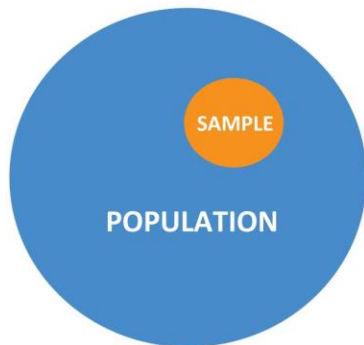
3. Sampling Methods

Sampling methods are techniques used to select samples from a population.

(i) Simple Random Sampling

Every element has an equal chance of selection.

👉 Example: Lottery method



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(ii) Stratified Sampling

Population is divided into groups (strata), and samples are taken from each group.

👉 Example: Students divided by class (CSE, ECE, etc.)

(iii) Systematic Sampling

Samples are selected at regular intervals.

👉 Example: Selecting every 10th student

(iv) Cluster Sampling

Population is divided into clusters, and entire clusters are selected randomly.

👉 Example: Selecting entire classrooms

(v) Convenience Sampling

Samples are chosen based on ease or availability.

👉 Example: Surveying nearby people

4. Conclusion

Sampling helps to study large populations efficiently with less time and cost.

Q28. Define estimation and write different types of estimations.

1. Estimation

Estimation is the **process of determining an approximate value of an unknown population parameter** using sample data.

2. Types of Estimation

(i) Point Estimation

It gives a **single value** as an estimate of the unknown parameter.

👉 Example:

Population mean = 50

👉 Note:

It may not exactly equal the true value

(ii) Interval Estimation

It gives a **range of values** within which the parameter is expected to lie.

👉 Example:

$45 < \mu < 55$

👉 Note:

More reliable than point estimation

3. Conclusion

Interval estimation is preferred because it provides a range and includes possible error.

8. Calculate the Mean of the following data

C.I	5–10	10–15	15–20	20–25	25–30	30–35	35–40
f	6	8	17	21	15	11	12

Step 1: Find Mid Values (x)

C.I	f	Mid (x)	f×x
5–10	6	7.5	45
10–15	8	12.5	100
15–20	17	17.5	297.5
20–25	21	22.5	472.5
25–30	15	27.5	412.5
30–35	11	32.5	357.5
35–40	12	37.5	450

Step 2: Find Totals

$$\Sigma f = 6 + 8 + 17 + 21 + 15 + 11 + 12 = \mathbf{90}$$

$$\Sigma fx = 45 + 100 + 297.5 + 472.5 + 412.5 + 357.5 + 450 = \mathbf{2135}$$

Step 3: Apply Formula

$$\text{Mean} = \Sigma fx / \Sigma f$$

$$= 2135 / 90$$

$$= \mathbf{23.72}$$

 **Final Answer**

$$\text{Mean} = \mathbf{23.72}$$

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9Q. Calculate the Median of the following data

C.I	5–10	10–15	15–20	20–25	25–30	30–35	35–40
f	6	8	17	21	15	11	12

Step 1: Find Cumulative Frequency (cf)

C.I	f	cf
5–10	6	6
10–15	8	14
15–20	17	31
20–25	21	52
25–30	15	67
30–35	11	78
35–40	12	90

👉 Total frequency:

$$N = \Sigma f = 90$$

Step 2: Find N/2

$$N/2 = 90 / 2 = 45$$

Step 3: Identify Median Class

👉 The cumulative frequency just greater than 45 is **52**

So, **Median Class = 20–25**

Step 4: Write Formula

$$\text{Median} = L + [(N/2 - cf) / f] \times h$$

Step 5: Identify Terms

- L = Lower limit of median class = **20**
- cf = Cumulative frequency of previous class = **31**
- f = Frequency of median class = **21**

- $h = \text{Class width} = 5$
-

Step 6: Substitute Values

$$\text{Median} = 20 + [(45 - 31) / 21] \times 5$$

Step 7: Solve

$$= 20 + (14 / 21) \times 5$$

$$= 20 + (70 / 21)$$

$$= 20 + 3.33$$

Final Answer

$$\text{Median} = 23.33$$

18 Q. Population = {3, 4, 5, 6, 7}

No. of elements, $N = 5$

Sample size, $n = 2$

Sampling: **With Replacement**

S-1: Number of samples

$$\text{No. of samples} = N^n = 5^2 = 25$$

S-2: Possible sample pairs

(3,3), (3,4), (3,5), (3,6), (3,7)

(4,3), (4,4), (4,5), (4,6), (4,7)

(5,3), (5,4), (5,5), (5,6), (5,7)

(6,3), (6,4), (6,5), (6,6), (6,7)

(7,3), (7,4), (7,5), (7,6), (7,7)

S-3: Mean of population

$$\mu_p = (3 + 4 + 5 + 6 + 7) / 5$$

$$= 25 / 5$$

👉 $\mu_p = 5$

S-4: SD of population

$$\sigma_p^2 = \Sigma(x - \mu_p)^2 / N$$

$$= [4 + 1 + 0 + 1 + 4] / 5$$

$$= 2$$

$$\sigma_p = \sqrt{2}$$

👉 $\sigma_p = 1.414$

S-5: Mean of sampling distribution

$$\mu_x^- = (\text{Sum of all sample means}) / 25$$

From pairs:

$\bar{x}$ values:

3, 3.5, 4, 4.5, 5

3.5, 4, 4.5, 5, 5.5

4, 4.5, 5, 5.5, 6

4.5, 5, 5.5, 6, 6.5

5, 5.5, 6, 6.5, 7

$$\Sigma \bar{x} = 125$$

$$\mu_x^- = 125 / 25$$

👉 $\mu_x^- = 5$

S-6: SD of sampling distribution (Manual)

$\bar{x}$	$(\bar{x} - 5)^2$	Frequency
3	4	1
3.5	2.25	2
4	1	3
4.5	0.25	4
5	0	5

5.5	0.25	4
6	1	3
6.5	2.25	2
7	4	1

Step calculation

$$\begin{aligned}
 & \Sigma f(\bar{x} - 5)^2 \\
 &= 4(1) + 2.25(2) + 1(3) + 0.25(4) + 0(5) + 0.25(4) + 1(3) + 2.25(2) + 4(1) \\
 &= 4 + 4.5 + 3 + 1 + 0 + 1 + 3 + 4.5 + 4 \\
 &= 25
 \end{aligned}$$

$$\sigma_x^2 = 25 / 25 = 1$$

$$\sigma_x = \sqrt{1}$$

$$\Rightarrow \sigma_x = 1$$

✓ Final Answers

1. Mean of population $\rightarrow 5$
2. SD of population $\rightarrow 1.414$
3. Mean of sampling distribution $\rightarrow 5$
4. SD of sampling distribution $\rightarrow 1$

Q. Population = {3, 4, 5, 6, 7}

No. of elements, $N = 5$

Sample size, $n = 2$

Sampling: **Without Replacement**

S-1: Number of samples

No. of samples $= {}^5C_2 = 10$

S-2: Possible sample pairs

(3,4), (3,5), (3,6), (3,7)
(4,5), (4,6), (4,7)
(5,6), (5,7)
(6,7)

S-3: Mean of population

$$\mu_p = (3 + 4 + 5 + 6 + 7) / 5$$
$$= 25 / 5$$

👉 $\mu_p = 5$

S-4: SD of population

$$\sigma_p^2 = \Sigma(x - \mu_p)^2 / N$$
$$= [4 + 1 + 0 + 1 + 4] / 5$$
$$= 2$$
$$\sigma_p = \sqrt{2}$$

👉 $\sigma_p = 1.414$

S-5: Mean of each sample

$$\bar{x} = (x_1 + x_2) / 2$$

Sample	$\bar{x}$
(3,4)	3.5
(3,5)	4
(3,6)	4.5
(3,7)	5
(4,5)	4.5
(4,6)	5
(4,7)	5.5
(5,6)	5.5
(5,7)	6
(6,7)	6.5

S-6: Mean of sampling distribution

$$\mu_{\bar{x}} = \Sigma \bar{x} / 10$$

$$\Sigma \bar{x} = 50$$

$$\mu_{\bar{x}} = 50 / 10$$

👉 $\mu_{\bar{x}} = 5$

S-7: SD of sampling distribution

$\bar{x}$	$(\bar{x} - 5)^2$
3.5	2.25
4	1
4.5	0.25
5	0
4.5	0.25
5	0
5.5	0.25
5.5	0.25
6	1
6.5	2.25

Calculation

$$\begin{aligned} & \Sigma(\bar{x} - 5)^2 \\ &= 2.25 + 1 + 0.25 + 0 + 0.25 + 0 + 0.25 + 0.25 + 1 + 2.25 \\ &= 7.5 \end{aligned}$$

$$\sigma_{\bar{x}}^2 = 7.5 / 10 = 0.75$$

$$\sigma_{\bar{x}} = \sqrt{0.75}$$

👉 $\sigma_{\bar{x}} = 0.866$

✅ Final Answers

1. Mean of population $\rightarrow 5$
2. SD of population $\rightarrow 1.414$
3. Mean of sampling distribution $\rightarrow 5$
4. SD of sampling distribution $\rightarrow 0.866$

20 Q. Population = {4, 8, 12, 16, 20, 24}

No. of elements, $N = 6$

Sample size, $n = 2$

Sampling: **Without Replacement**

S-1: Number of samples

No. of samples = ${}^6C_2 = 15$

S-2: Possible sample pairs

(4,8), (4,12), (4,16), (4,20), (4,24)
(8,12), (8,16), (8,20), (8,24)
(12,16), (12,20), (12,24)
(16,20), (16,24)
(20,24)

S-3: Mean of population

$$\mu_p = (4 + 8 + 12 + 16 + 20 + 24) / 6$$

$$= 84 / 6$$

👉 $\mu_p = 14$

S-4: SD of population

x	$(x - 14)^2$
4	100
8	36
12	4

16	4
20	36
24	100

$$\Sigma = 280$$

$$\sigma_p^2 = 280 / 6 = 46.67$$

$$\sigma_p = \sqrt{46.67}$$

👉 $\sigma_p = 6.83$

S-5: Mean of each sample

$$\bar{x} = (x_1 + x_2) / 2$$

Sample	$\bar{x}$
(4,8)	6
(4,12)	8
(4,16)	10
(4,20)	12
(4,24)	14
(8,12)	10
(8,16)	12
(8,20)	14
(8,24)	16
(12,16)	14
(12,20)	16
(12,24)	18
(16,20)	18
(16,24)	20
(20,24)	22

S-6: Mean of sampling distribution

$$\Sigma \bar{x} = 210$$

$$\mu_x^- = 210 / 15$$

👉 $\mu_x^- = 14$

S-7: SD of sampling distribution

$\bar{x}$	$(\bar{x} - 14)^2$
6	64
8	36
10	16
12	4
14	0
10	16
12	4
14	0
16	4
14	0
16	4
18	16
18	16
20	36
22	64

Calculation

$$\begin{aligned}\Sigma(\bar{x} - 14)^2 \\ = 280\end{aligned}$$

$$\sigma_x^2 = 280 / 15 = 18.67$$

$$\sigma_x^- = \sqrt{18.67}$$

👉 $\sigma_x^- = 4.32$

✓ **Final Answers**

1. Mean of population → **14**
2. SD of population → **6.83**
3. Mean of sampling distribution → **14**
4. SD of sampling distribution → **4.32**